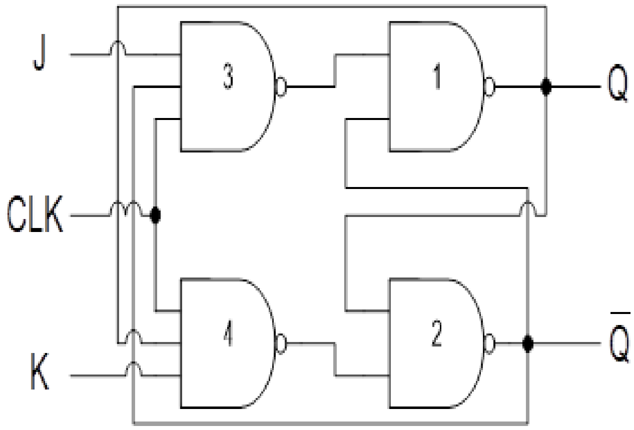
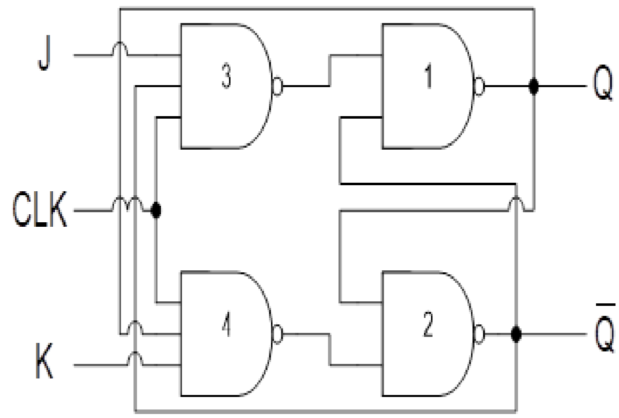
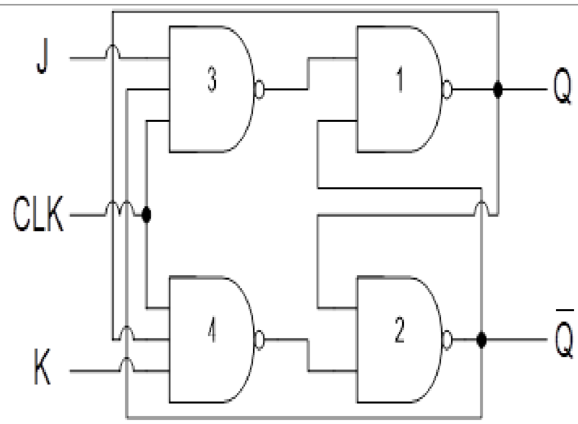
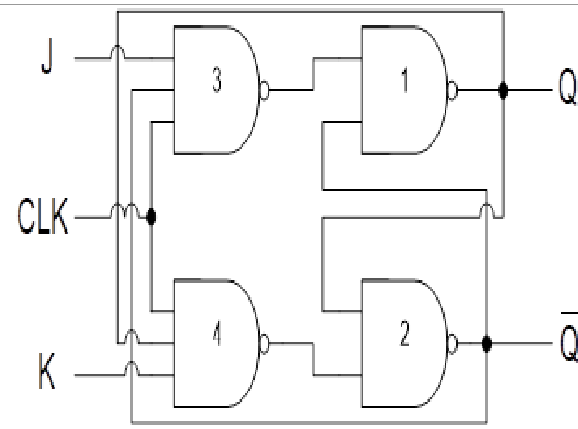
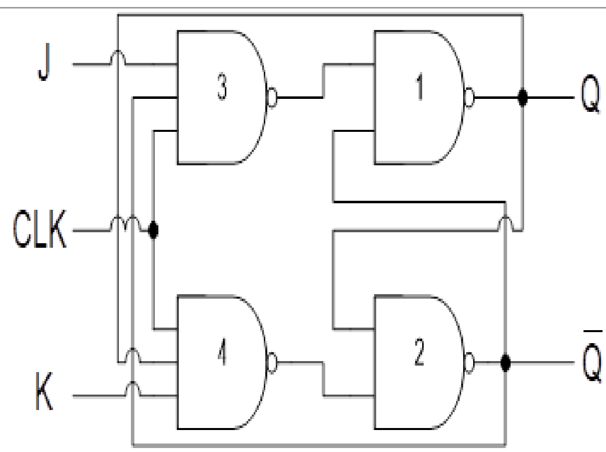
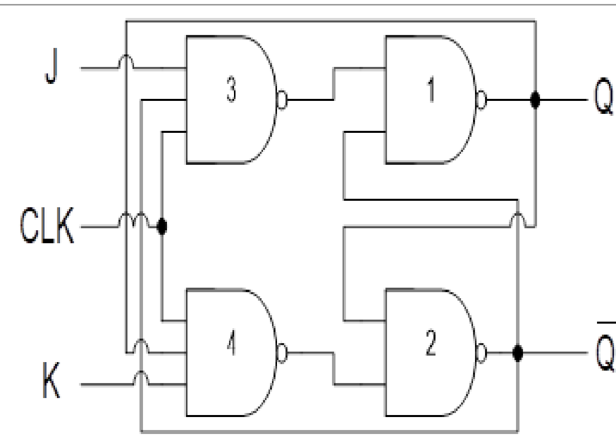
QUIZ 7-A

ID:________ NAME:__________________________________ SECTION: ____

Instruction: **No marks will be awarded for Direct Answers**

1. *Process the **JK LATCH** given below for all possible cases and fill the table accordingly?*

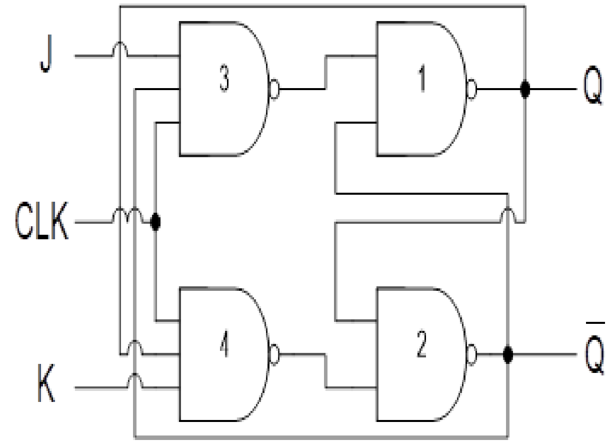
CIRCUIT:

<table>
<tr><td>CLK=0, J= , K= , Q_{T+1} = $\overline{Q_{T+1}}$ =</td><td>CLK=0, J= , K= , Q_{T+1} = $\overline{Q_{T+1}}$ =</td></tr>
<tr><td></td><td></td></tr>
<tr><td>CASE I
($Q_{t=}$ 0)</td><td>CASE-II
($Q_{t=}$ 1)</td></tr>
<tr><td>CLK=1, J=0 , K=0, Q_{T+1} = $\overline{Q_{T+1}}$ =</td><td>CLK=1, J=0 , K=0, Q_{T+1} = $\overline{Q_{T+1}}$ =</td></tr>
<tr><td></td><td></td></tr>
<tr><td>CLK=1, J=0 , K=1, Q_{T+1} = $\overline{Q_{T+1}}$ =</td><td>CLK=1, J=0 , K=1, Q_{T+1} = $\overline{Q_{T+1}}$ =</td></tr>
<tr><td></td><td></td></tr>
</table>

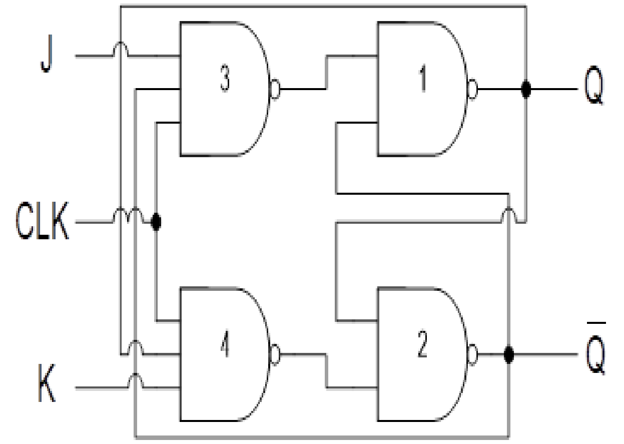
QUIZ 7-A

ID:________ NAME:________________________________ SECTION: ____

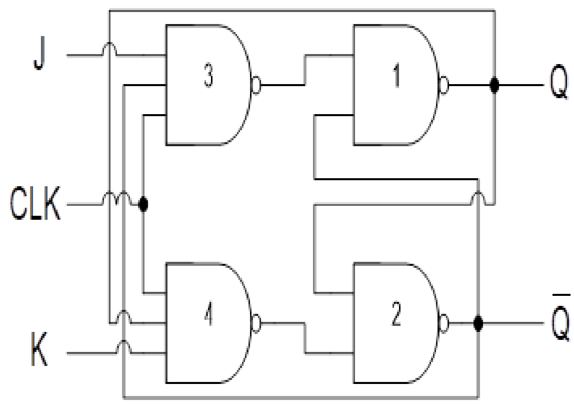
CLK=1, J=1 , K=1, Q_{T+1} = $\overline{Q_{T+1}}$ =



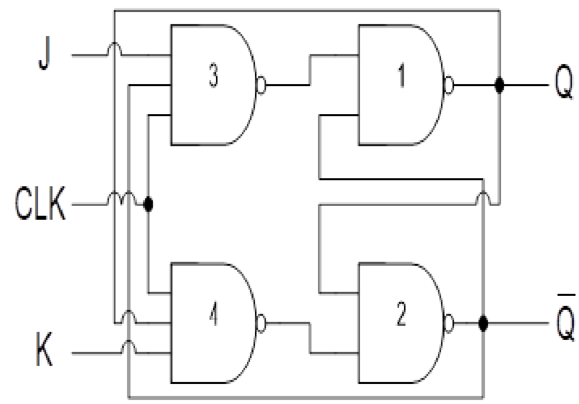
CLK=1, J=0 , K=1, Q_{T+1} = $\overline{Q_{T+1}}$ =



CLK=1, J=0 , K=1, Q_{T+1} = $\overline{Q_{T+1}}$ =



CLK=1, J=0 , K=1, Q_{T+1} = $\overline{Q_{T+1}}$ =



JK Latch

E	J	K	Q_{t+1}	$\overline{Q_{t+1}}$
0	X	X		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

JK LATCH with SET/RESET

SET	CLEAR	Q_{t+1}	$\overline{Q}_{t+1}$
0	0		
0	1		
1	0		
1	1		

2. *Modify given **NAND** Based JK FlipFlop circuit to introduce Asynchronous **CLEAR** and **SET** inputs fill table above?*